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PERBANDINGAN ALGORITMA

LOGISTIC REGRESSION, RANDOM FOREST DAN XGBOOST UNTUK

KLASIFIKASI PERSETUJUAN PINJAMAN BANK

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Abstraksi

Studi ini menguji secara komparatif performa Logistic Regression, Random Forest, dan XGBoost dalam memprediksi persetujuan pinjaman bank. Kami melakukan penelitian ini melalui serangkaian tahapan: pra-pemrosesan data, seleksi fitur, pembagian data, pelatihan model, hingga evaluasi menyeluruh menggunakan metrik akurasi, presisi, recall, dan F1-score. Dataset yang kami gunakan tersusun atas enam fitur krusial, meliputi jumlah pinjaman, suku bunga, pendapatan, kepemilikan rumah, tujuan pinjaman, dan riwayat gagal bayar. Secara signifikan, XGBoost terbukti memberikan akurasi tertinggi di antara algoritma yang diuji. Penemuan ini diharapkan dapat memfasilitasi lembaga keuangan untuk mengotomatisasi dan meningkatkan efisiensi proses seleksi peminjam mereka.

Kata Kunci:

klasifikasi, persetujuan pinjaman, logistic regression, random forest, xgboost

Abstract

This study comprehensively compares the performance of three classification algorithms: Logistic Regression, Random Forest, and XGBoost in predicting bank loan approvals. We conducted the research through stages of data preprocessing, feature selection, data splitting, model training, and a thorough evaluation using accuracy, precision, recall, and F1-score metrics. The dataset utilized in this work comprises six crucial features: loan amount, interest rate, income, home ownership, loan intent, and previous default history. Significantly, XGBoost demonstrated the highest accuracy among the tested algorithms. This discovery is anticipated to empower financial institutions to automate and enhance the efficiency of their borrower selection processes.

Keywords:

classification, loan approval, logistic regression, random forest, xgboost

Pendahuluan

Dalam lanskap keuangan modern, melonjaknya permintaan pinjaman dari masyarakat mendesak lembaga finansial untuk mengimplementasikan proses seleksi calon peminjam yang presisi dan efisien. Kemampuan memprediksi kelayakan peminjam menjadi sangat krusial demi menjaga stabilitas keuangan bank dan memitigasi risiko gagal bayar. Oleh karena itu, penerapan metode klasifikasi dalam domain data mining menjadi esensial untuk menganalisis data peminjam yang tersedia, sehingga memungkinkan prediksi persetujuan pinjaman secara otomatis dan akurat.

Tinjauan literatur menunjukkan bahwa algoritma machine learning telah terbukti efektif dalam memprediksi risiko kredit dan keputusan pinjaman [1], [2]. Secara khusus, Random Forest telah menunjukkan konsistensi dalam memberikan hasil prediksi yang stabil dan kuat, bahkan pada dataset dengan fitur yang melimpah dalam sektor keuangan [5], [7]. Di sisi lain, XGBoost menonjol karena kapabilitasnya dalam menangani ketidakseimbangan data dan interaksi kompleks antar fitur, seringkali menghasilkan kinerja superior dalam tugas prediksi pinjaman [3], [9]. Sementara itu, Logistic Regression, sebagai fondasi model prediktif, tetap relevan berkat kemampuannya memberikan interpretasi yang jelas dalam penilaian risiko kredit [4].

Melanjutkan studi-studi terdahulu, penelitian ini mengambil langkah lebih jauh dengan membandingkan performa tiga algoritma klasifikasi Logistic Regression, Random Forest, dan XGBoost dari segi akurasi prediksi pada dataset persetujuan pinjaman. Pendekatan komparatif ini sejalan dengan upaya-upaya serupa dalam literatur prediksi risiko kredit [1], [2], [7], namun dengan fokus spesifik pada aplikasi dan perbandingan langsung algoritma-algoritma tersebut dalam konteks persetujuan pinjaman.

Metode Penelitian

Riset ini menjalankan serangkaian tahapan metodologis yang sistematis guna mencapai tujuannya. Prosesnya dimulai dengan pengumpulan data, diikuti pra-pemrosesan yang cermat. Selanjutnya, kami melakukan pemilihan fitur esensial berdasarkan analisis korelasi yang divisualisasikan menggunakan heatmap, sebuah langkah krusial untuk mengidentifikasi variabel-variabel relevan. Setelah itu, tahapan inti berlanjut pada pelatihan model-model klasifikasi yang diusulkan: Logistic Regression, Random Forest, dan XGBoost. Terakhir, performa model dievaluasi secara komprehensif untuk memprediksi persetujuan pinjaman bank. Seluruh pendekatan ini dirancang untuk membangun model klasifikasi yang efektif dan optimal dalam konteks prediksi tersebut.

Diagram Alir Penelitian

Diagram alir berikut menggambarkan tahapan penelitian secara keseluruhan:

Gambar 2.1 Diagram Alir Penelitian

Penjelasan Tahapan Penelitian

Untuk mencapai tujuan penelitian, studi ini menjalankan serangkaian tahapan metodologis krusial yang dirancang untuk membangun model klasifikasi persetujuan pinjaman bank yang akurat dan sistematis. Rincian tahapan-tahapan ini disajikan pada sub-bagian berikut:

Peng

● Sentences that are likely AI-generated.

FAQs

What is GPTZero?

GPTZero is the leading AI detector for checking whether a document was written by a large language model such as ChatGPT. GPTZero detects AI on sentence, paragraph, and document level. Our model was trained on a large, diverse corpus of human-written and AI-generated text, with a focus on English prose. To date, GPTZero has served over 10 million users around the world, and works with over 100 organizations in education, hiring, publishing, legal, and more.

When should I use GPTZero?

Our users have seen the use of AI-generated text proliferate into education, certification, hiring and recruitment, social writing platforms, disinformation, and beyond. We've created GPTZero as a tool to highlight the possible use of AI in writing text. In particular, we focus on classifying AI use in prose. Overall, our classifier is intended to be used to flag situations in which a conversation can be started (for example, between educators and students) to drive further inquiry and spread awareness of the risks of using AI in written work.

Does GPTZero only detect ChatGPT outputs?

No, GPTZero works robustly across a range of AI language models, including but not limited to ChatGPT, GPT-4, GPT-3, GPT-2, LLaMA, and AI services based on those models.

What are the limitations of the classifier?

The nature of AI-generated content is changing constantly. As such, these results should not be used to punish students. We recommend educators to use our behind-the-scenes [Writing Reports](#) as part of a holistic assessment of student work. There always exist edge cases with both instances where AI is classified as human, and human is classified as AI. Instead, we recommend educators take approaches that give students the opportunity to demonstrate their understanding in a controlled environment and craft assignments that cannot be solved with AI. Our classifier is not trained to identify AI-generated text after it has been heavily modified after generation (although we estimate this is a minority of the uses for AI-generation at the moment). Currently, our classifier can sometimes flag other machine-generated or highly procedural text as AI-generated, and as such, should be used on more descriptive portions of text.

I'm an educator who has found AI-generated text by my students. What do I do?

Firstly, at GPTZero, we don't believe that any AI detector is perfect. There always exist edge cases with both instances where AI is classified as human, and human is classified as AI. Nonetheless, we recommend that educators can do the following when they get a positive detection: Ask students to demonstrate their understanding in a controlled environment, whether that is through an in-person assessment, or through an editor that can track their edit history (for instance, using our [Writing Reports](#) through Google Docs). Check out our list of [several recommendations](#) on types of assignments that are difficult to solve with AI.

Ask the student if they can produce artifacts of their writing process, whether it is drafts, revision histories, or brainstorming notes. For example, if the editor they used to write the text has an edit history (such as Google Docs), and it was typed out with several edits over a reasonable period of time, it is likely the student work is authentic. You can use GPTZero's Writing Reports to replay the student's writing process, and view signals that indicate the authenticity of the work.

See if there is a history of AI-generated text in the student's work. We recommend looking for a long-term pattern of AI use, as opposed to a single instance, in order to determine whether the student is using AI.